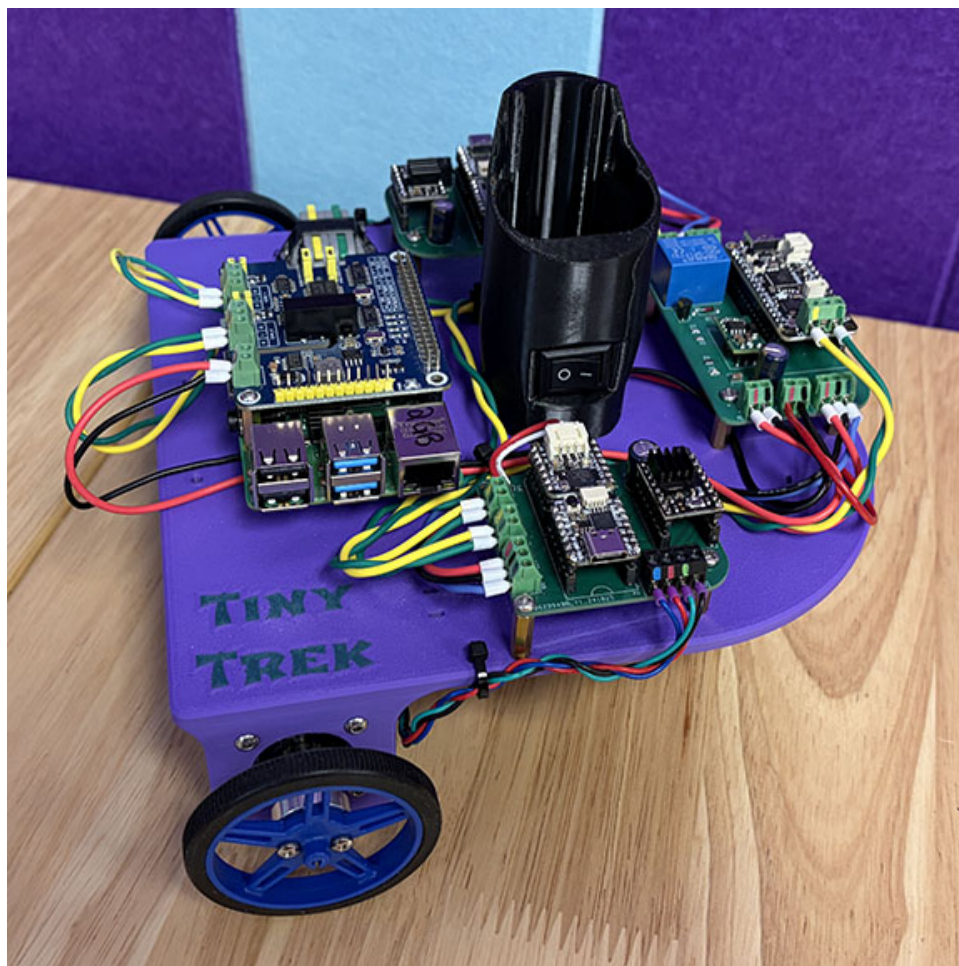


TinyTrek V1.5 Build Guide

Proto Squirrel

August 2026

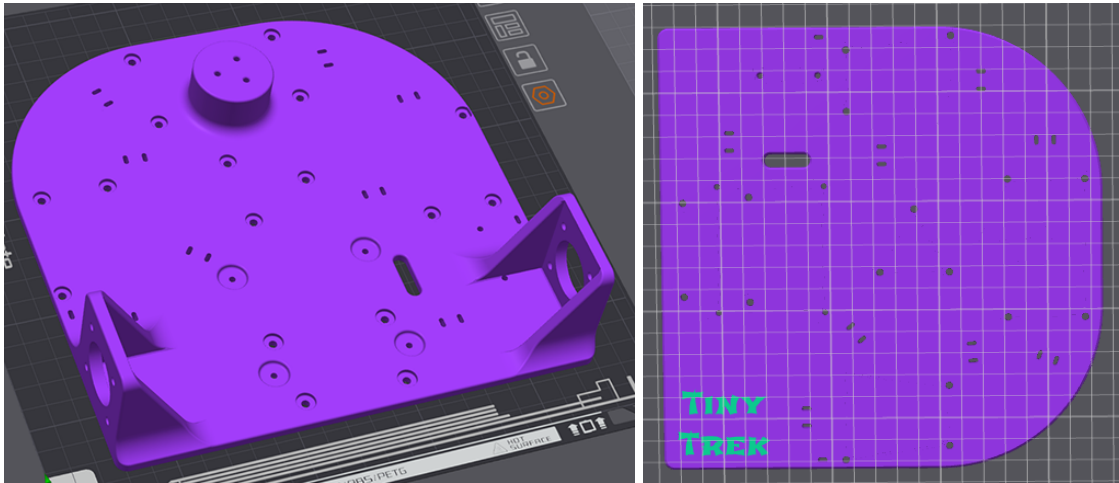


3D Print

There are three parts that must be printed to make the main Tiny Trek chassis and few optional parts. The print settings specified below are the settings we used to print the parts. They may not be the optimal settings for your machine.

Required parts

Chassis



You will need to print the chassis top-down on the build plate. The chassis is 8" x 8", so you will need a medium sized printer. There is an outline of a Tiny Trek logo that could be filled in for optional multi color printing. Textured build plates, such as PEI, work without issues.

Filament: We recommend using a higher-temp filament like HTPLA or PETG. Both work well in our testing. Something like regular PLA will probably work fine too, but ultimately that's up to you and how it fits your application.

Print settings:

Nozzle: 0.4mm

Layer Height: 0.2mm

Wall Loops: 3

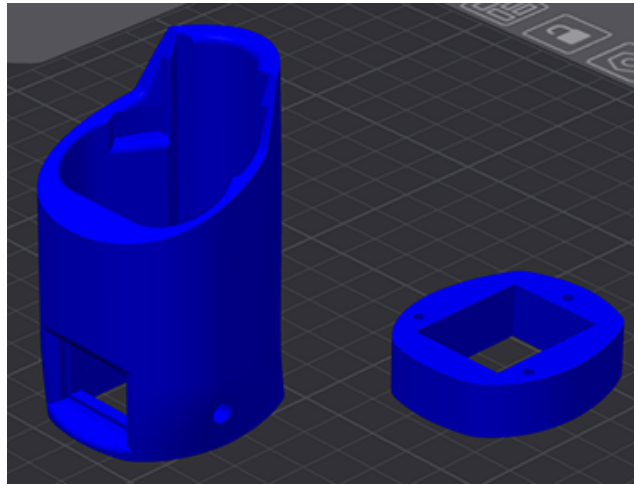
Top Shell Layers: 5

Bottom Shell Layers: 3

Infill: 20% Cubic

Supports: Not needed

Battery Receptacle



The other two required pieces are the M12 battery receptacle and the battery receptacle extender. The cavity in the extender provides room for the wiring after installing the switch. We used a model from an external source for the battery holder and added the extender to suit TinyTrek.

Filament: PLA and PETG both work, but given shrinkage and mechanical “slipperiness”, PETG is the superior choice for the battery receptacle. It tends to be easier to get the battery in and out.

Print Settings:

Nozzle: 0.4mm

Layer Height: 0.2mm

Wall Loops: 2

Top Shell Layers: 5

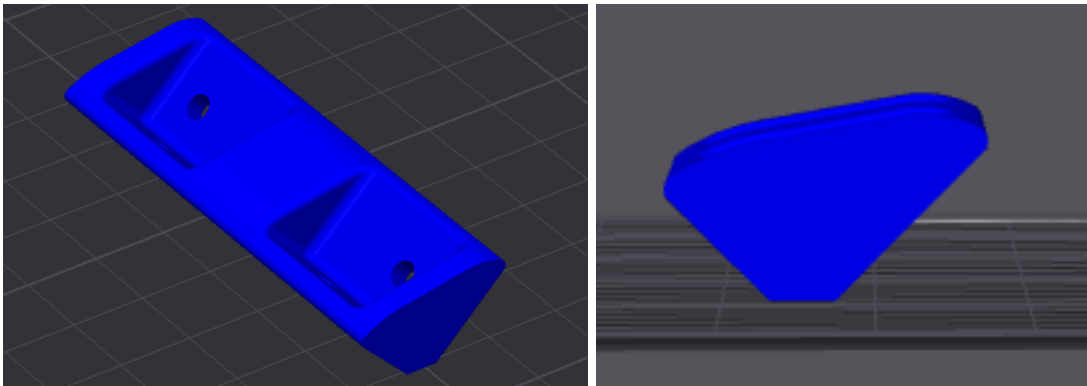
Bottom Shell Layers: 3

Infill: 15% Cubic

Supports: Difficult to remove, cleanup will be necessary whether you use supports or not given the way this particular model was designed.

Optional parts

Camera Bracket



This bracket allows you to mount DFRobot’s ESP32-S3 AI Camera Module in a vertical orientation at the front of the chassis giving a FPV of sorts. The bracket was designed with a chamfer so the print should be oriented in such a way that both ”legs” of the L shape are oriented at a 45 degree angle to the print bed.

Filament: This holds a potentially fragile part in a potentially precarious orientation. Stronger may be better. We used PETG.

Print settings:

Nozzle: 0.4mm

Layer Height: 0.2mm

Wall Loops: 3

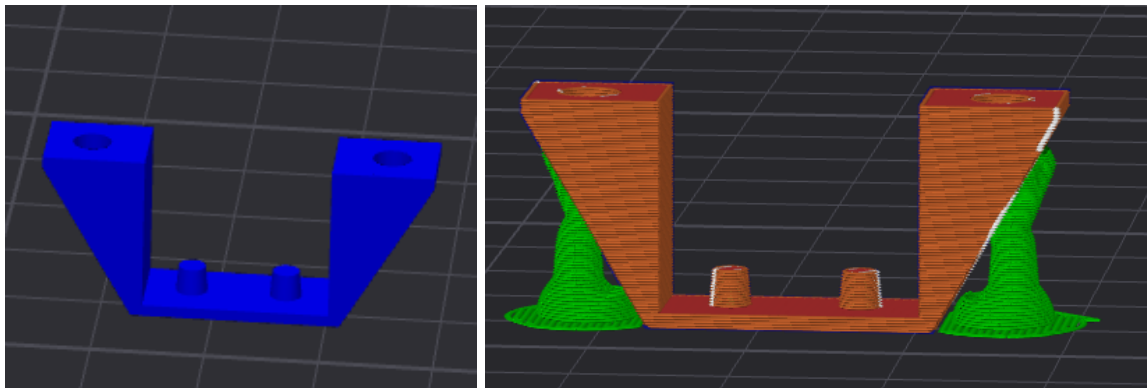
Top Shell Layers: 5

Bottom Shell Layers: 3

Infill: 20% Cubic

Supports: Not needed

Snap Connector Bracket



This bracket allows you to connect a pair of the recommended snap connectors to the chassis to provide a quick-release of an external device to one of the on-board CAN Buses. This bracket was specifically designed to slot into holes in the top of the specified connectors, so it may not work with other brands. Bracket should be printed upside down with the legs and small nubs facing up.

Print settings:

Nozzle: 0.4mm

Layer Height: 0.2mm

Wall Loops: 3

Top Shell Layers: 5

Bottom Shell Layers: 3

Infill: 20% Cubic

Supports: Tree supports are preferred to prevent drooping, but the part is so small that the drooping does not keep it from functioning.

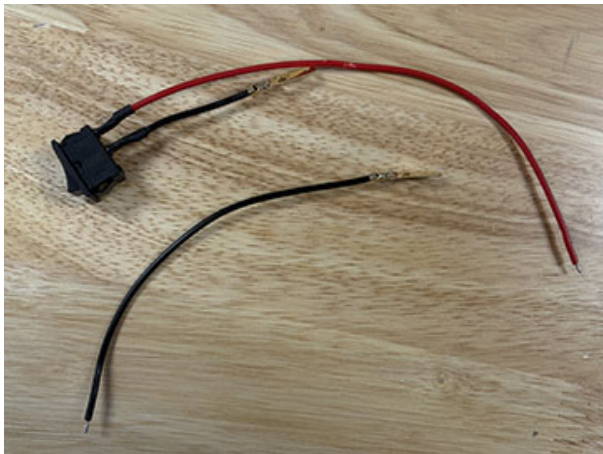
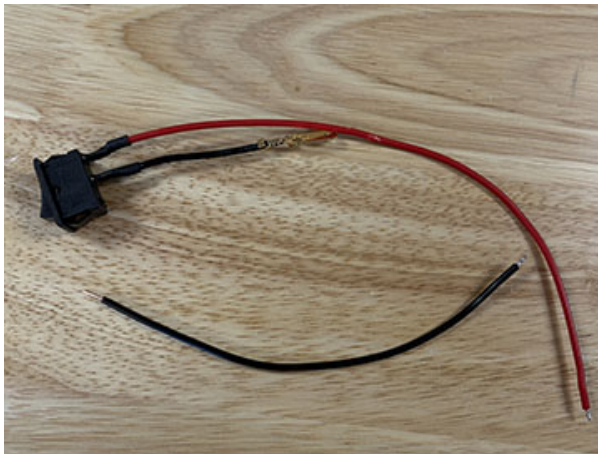
Assemble the Battery Receptacle

Parts:

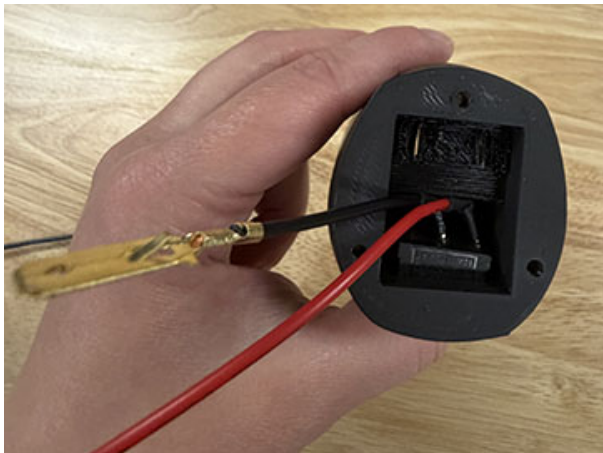
- 1x 3D Printed M12 Receptacle
- 1x 3D Printed Battery Receptacle Extender
- 2x 6.3mm Male Spade Connectors
- 1x power switch
- 3x M3x10mm screws (Optional)

Tools:

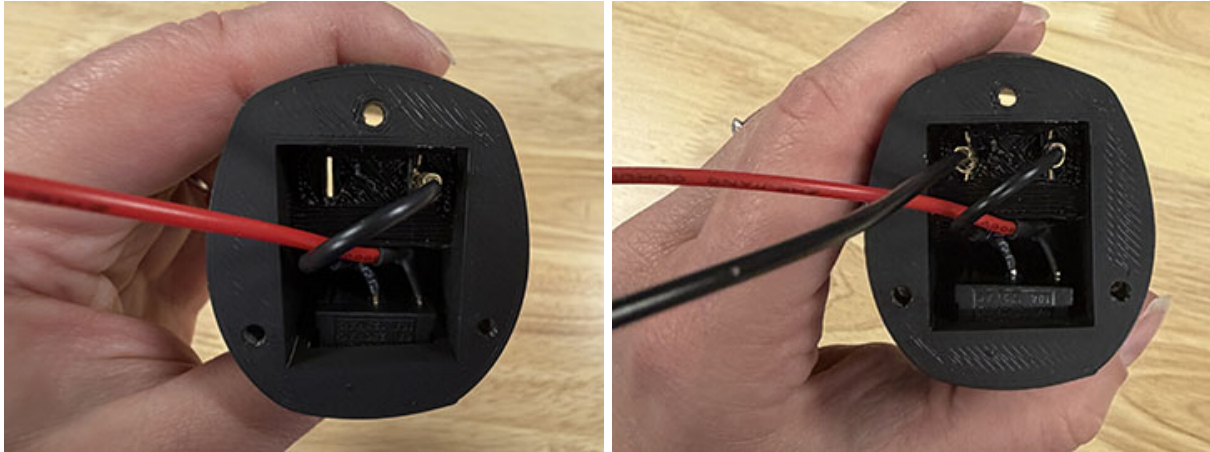
- hot glue gun and glue sticks (Optional)
- CA glue (Optional)



Cut the black wire on the switch leaving approximately 2 inches on the switch. Strip the two new ends of the black wire and crimp a male spade connector onto each of the newly stripped ends.



Insert the switch into the housing as shown above.



Cross the black wire to the other side of the housing and insert the spade connector into the slot on the right. Insert the other black wire with the spade connector into the left slot. The switch should now be on the hot side of the battery. Double check your connection by carefully inserting a battery into the receptacle and measuring the voltage at the wire ends both with the switch off and on. If you read a positive voltage with the battery, your receptacle is wired up properly.



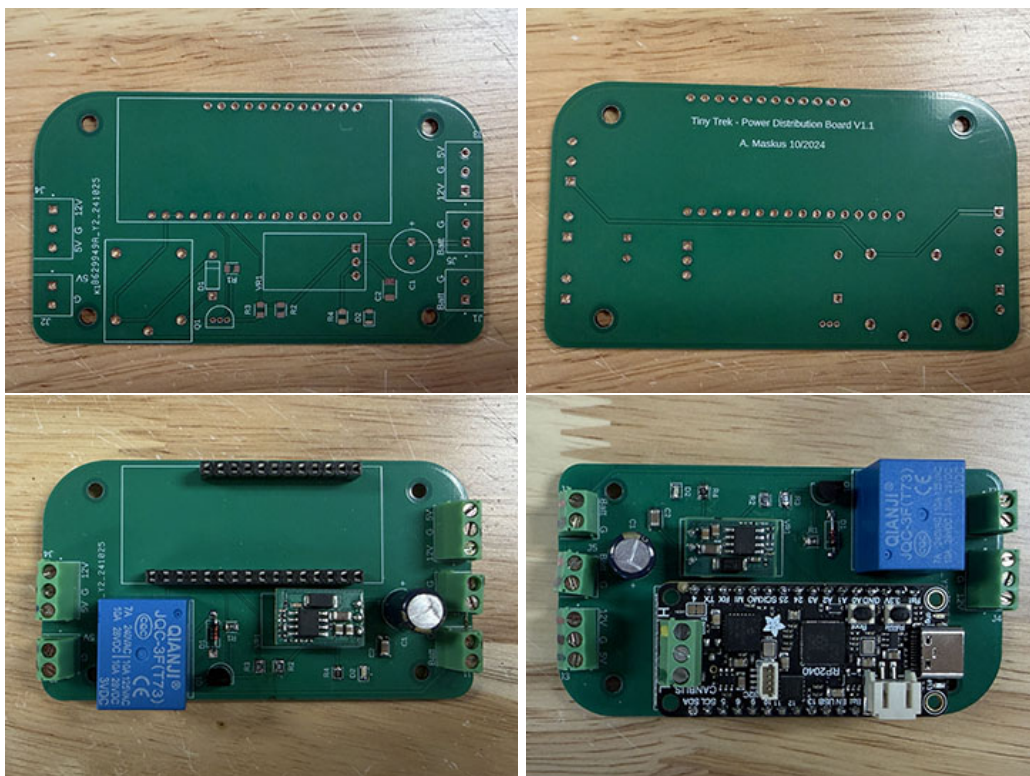
Optionally, you can reinforce the connection with hot glue to keep the spade connectors in place and add a small amount of strain resistance.



Feed the exposed wire ends through the small hole in the side of the extender and make sure the hole lines up on the left side of the receptacle. Attach the extender to the bottom of the receptacle with either 3 M3x10mm screws going in from the battery opening or by applying CA glue between the two pieces.

Assemble Custom PCBs

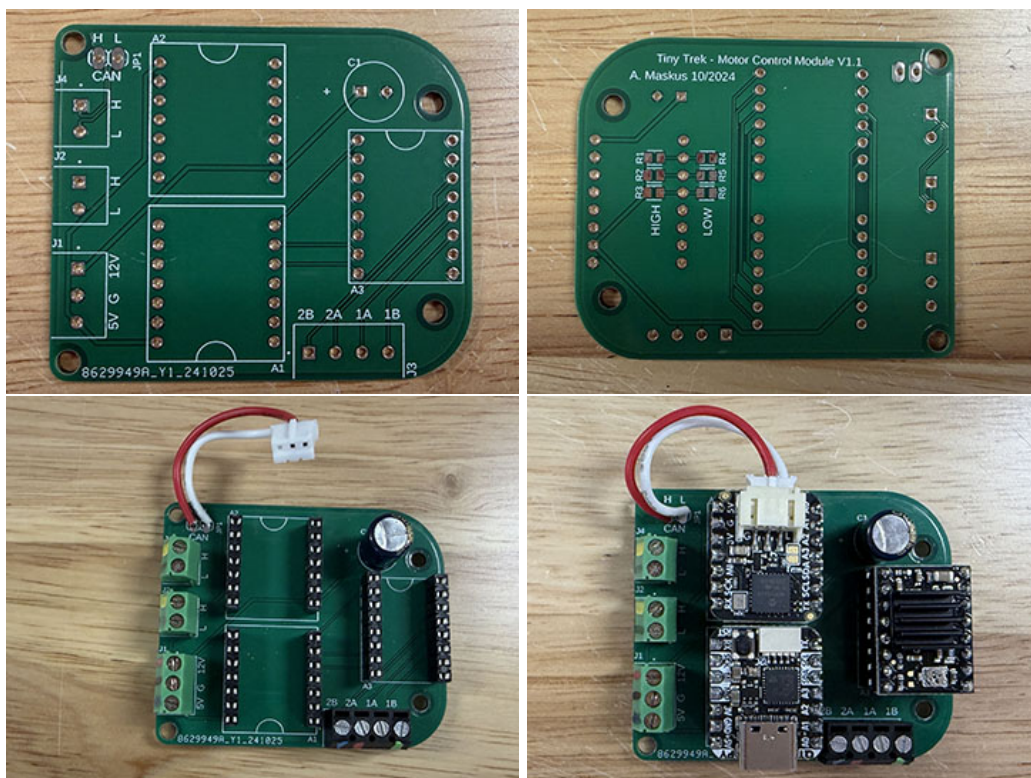
Assemble Power Distribution Board V1.1



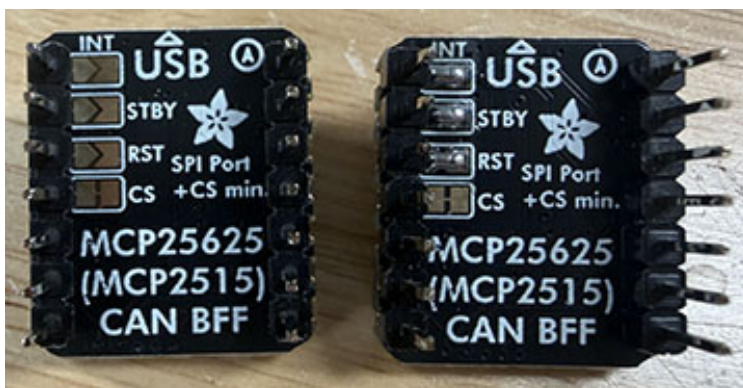
Assemble the Power Distribution Board as shown above.

A few things of note: There are a handfull of very similar Voltage regulators, but they aren't identical - check orientation against the proper datasheet. Input Voltage should go in the hole with the square pad for VR1. Headers for the Feather make it much easier to both program the Feather and swap out a misbehaving Feather.

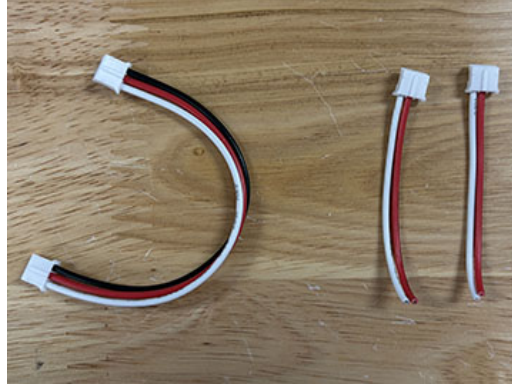
Assemble Motor Controller Boards



The two motor controller boards are assembled identically. The difference comes in the code and the way the motor coils are hooked up. Female headers in the board allow easy removal of the daughter boards here as well for programming the QTPY or replacing misbehaving parts.



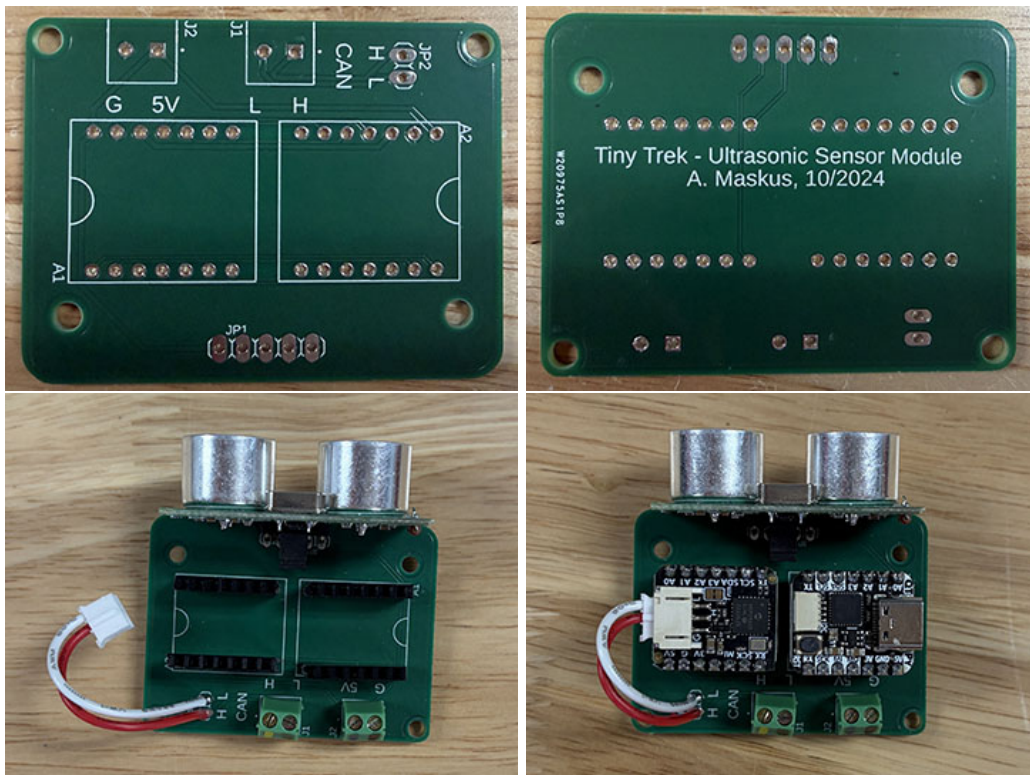
On the bottom side of the CANBFF there is a series of 4 jumpers. By default, INT, STBY and RST are open and CS is closed [left]. We want all four closed. This can be achieved with a solder blob [right]. I recommend making this modification before soldering headers to the CAN BFF. You will need to make this modification for both motor controllers and for the optional Ultrasonic Sensor module. These jumpers allow the CAN BFF to talk to the QTPY over SPI.



The JST pigtail is created by cutting the specified cable in half and removing the black wire. The resulting pigtail should have the white wire in the CAN-Low position and the red wire in the CAN-High position. Check the label on the Motor Controller Module board against the labels on the JST jack on the CAN BFF.

OPTIONAL: Assemble the Ultrasonic Sensor Module

CAUTION: We do not currently have any code written for the Ultrasonic Sensor Module, thus it has not been successfully tested. There is also a hardware jumper on the back of the US-100 Ultrasonic Sensor that is used to change the communication mode and may be impossible or at least incredibly difficult to change if you solder the ultrasonic sensor directly to the custom board. You may wish to use a female header there as well.

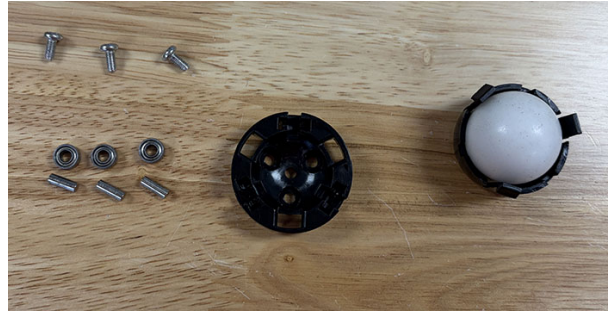


The Ultrasonic Sensor Module uses the same QTPY and CANBFF setup as the Motor Controller Modules, so the same caveats about the Jumpers on the back of the CANBFF and the CAN pigtail apply here as well.

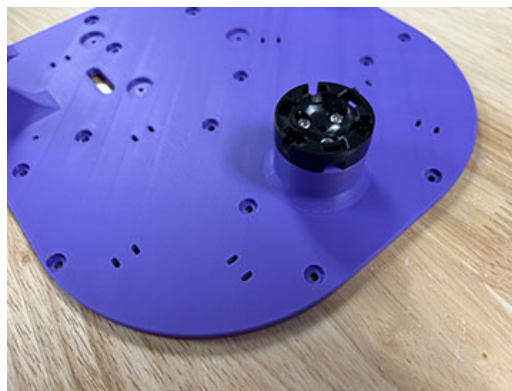
Mount Ball Caster

Parts:

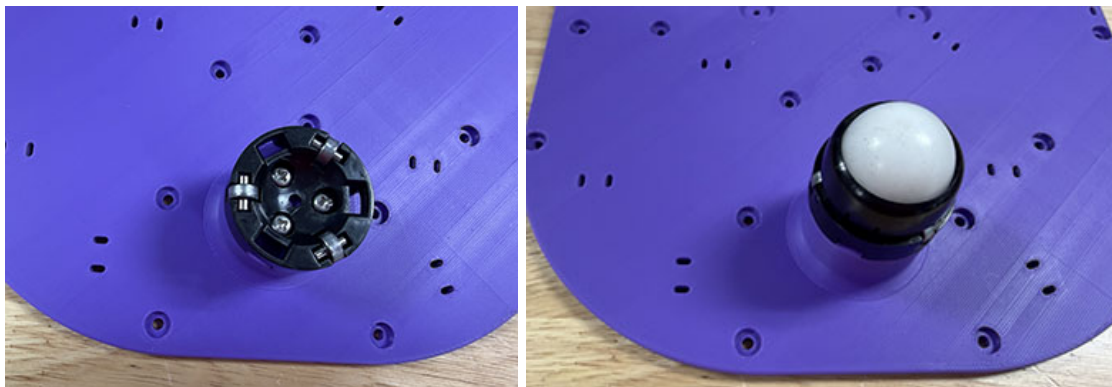
- Pololu Ball Caster kit
- 3 x M3x6mm screws



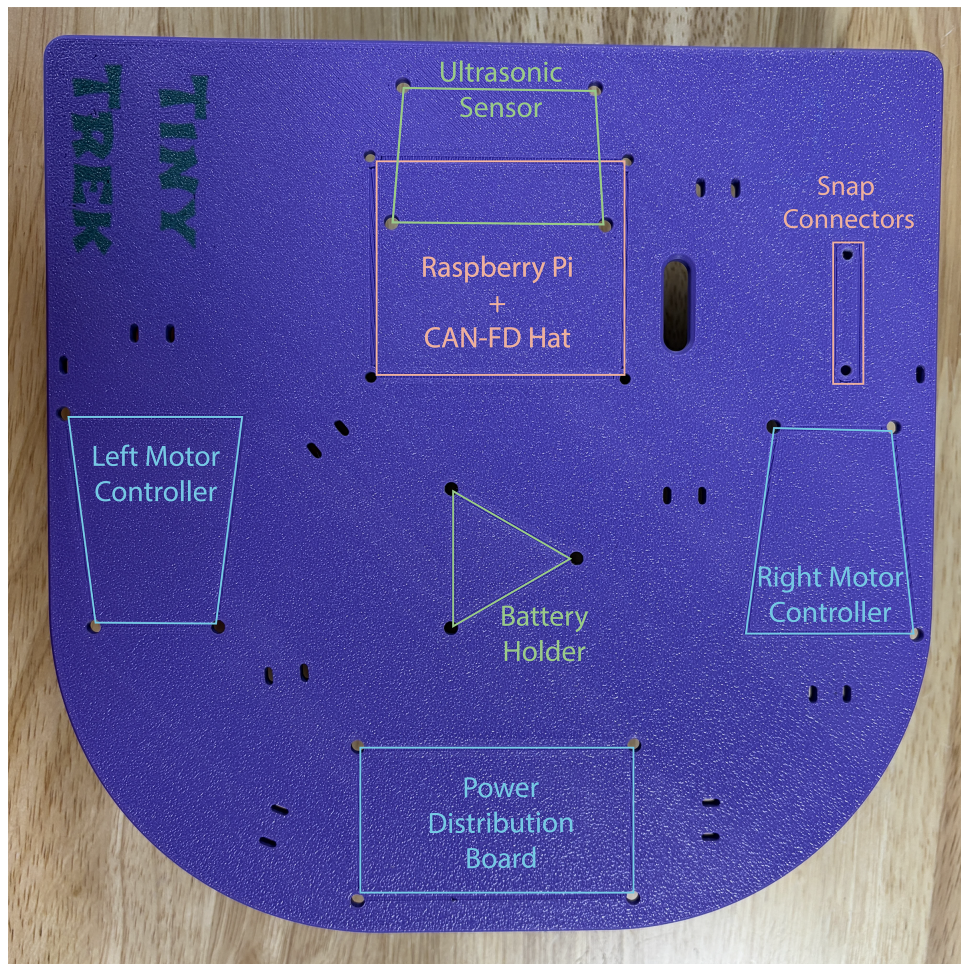
Before assembling the ball caster housing, screw the base to the chassis on the protrusion at the bottom back of the chassis with 3 M3x6mm screws.



Assemble the ball caster in place.



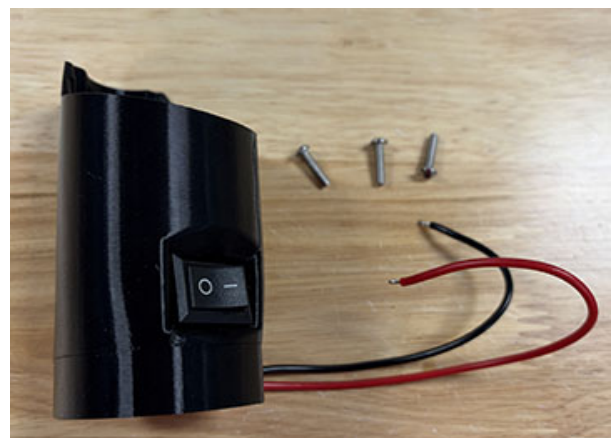
Mount Electronics to Chassis



Battery Receptacle

Parts:

- 1 x M12 Receptacle Assembly
- 3 x M3x12mm screws

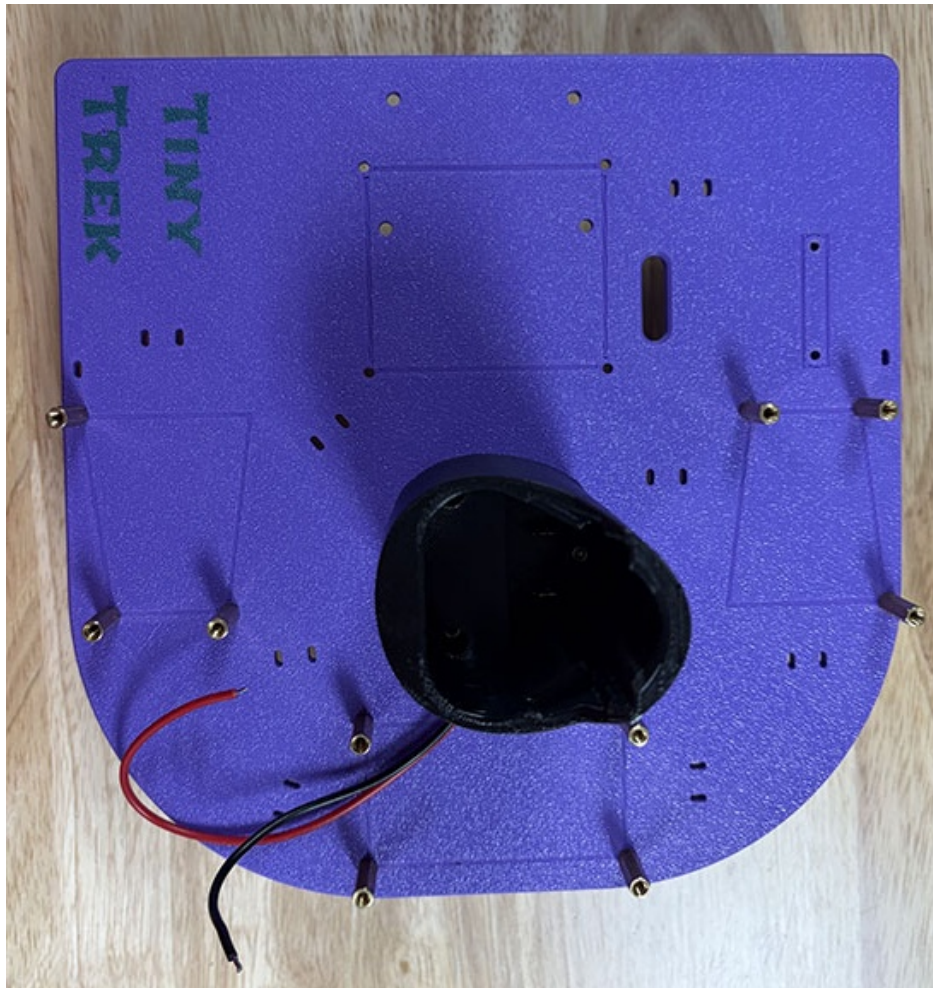


The battery receptacle assembly can be attached to the chassis on the triangle with 3 M3x12mm screws coming up from the bottom. The power switch will be facing the left side of the chassis, and the wires should be leading toward the rear of the chassis.

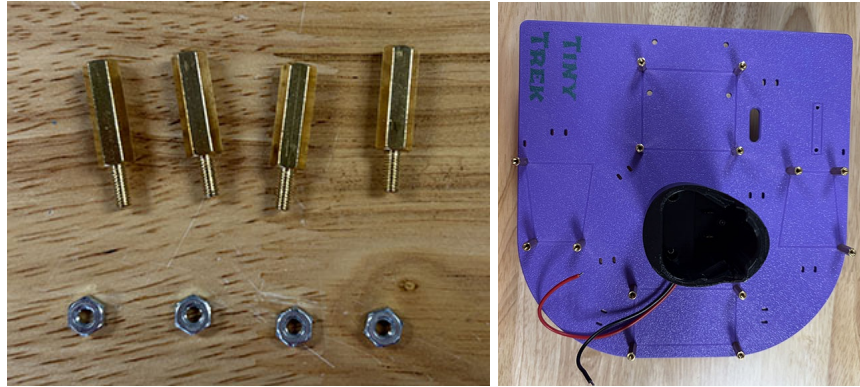
Standoffs

Parts:

- 10 x M3x6mm screws
- 10 x M3x16mm standoffs
- 4 x M2.5x15mm standoffs
- 4 x M2.5 nuts



Install M3x16mm standoffs on all holes on the three polygons outlined in blue above (for the motor boards and power distribution board) with standoffs on top, and M3x6mm screws on bottom.



Install M2.5 Standoffs on the 4 holes on the large rectangle for the Raspberry Pi with standoffs on top and nuts on the bottom of the chassis.

Optional Ultrasonic Sensor and/or Camera Board

Parts:

- 4 x M3x6mm screws
- 6 x M3x8mm Standoffs (16mm standoffs are ok, but not optimal here.)
- 1 x printed Camera Bracket
- 2 x M3x8mm screws

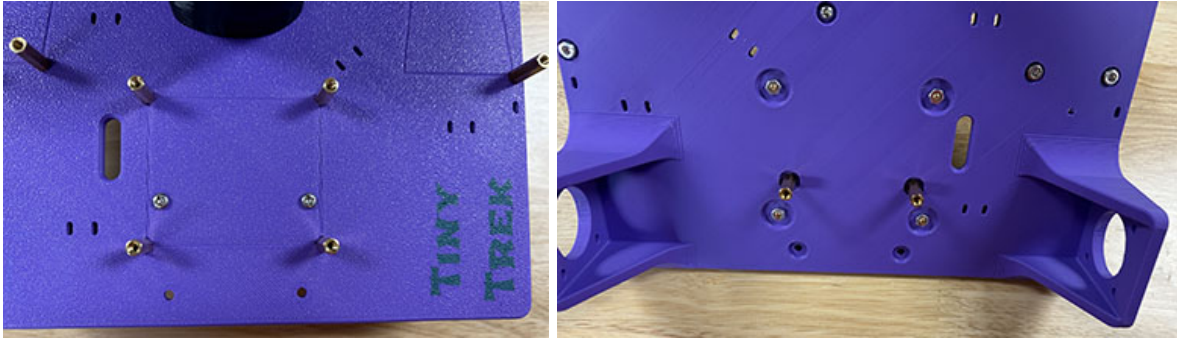


Camera Bracket



Install 2 M3x8mm standoffs on the side on the holes closer together. Standoffs on the flat side, M3x6mm screws on the inside of the bracket.

Ultrasonic Sensor Standoffs



Install M3x8mm standoffs in the back 2 holes with standoffs on bottom, M3x6mm screws on top.



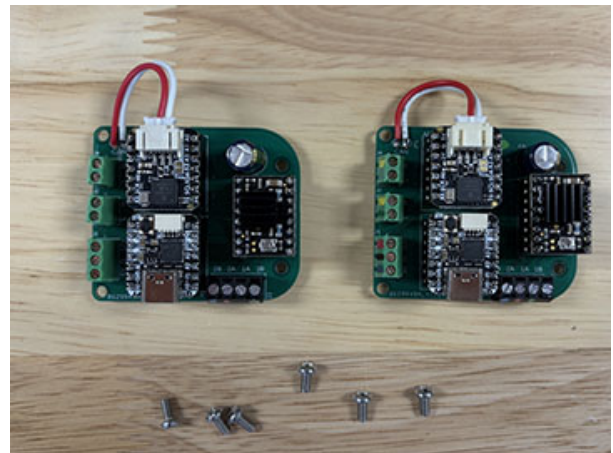
If using the Camera Bracket, 2 M3x8mm screws go through the Bracket and the Chassis into the standoffs at the front of the chassis.

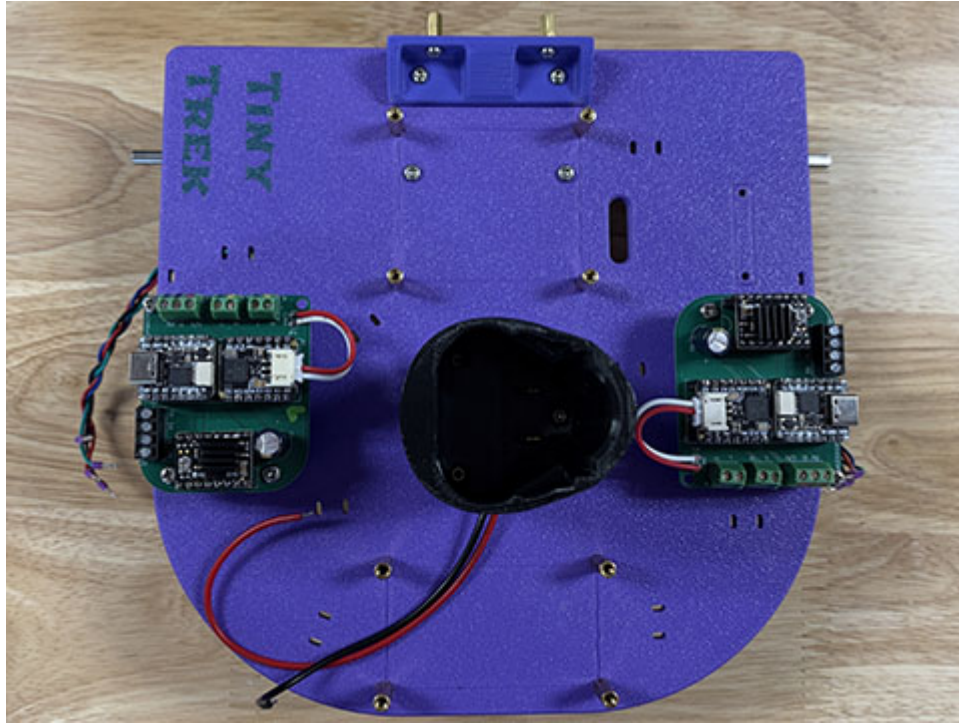
(If you aren't using the camera, the front two standoffs install the same way as the back two.)

Motor controller boards

Parts:

- 2 x Assembled Motor Controller Boards
- 6 x M3x6mm screws



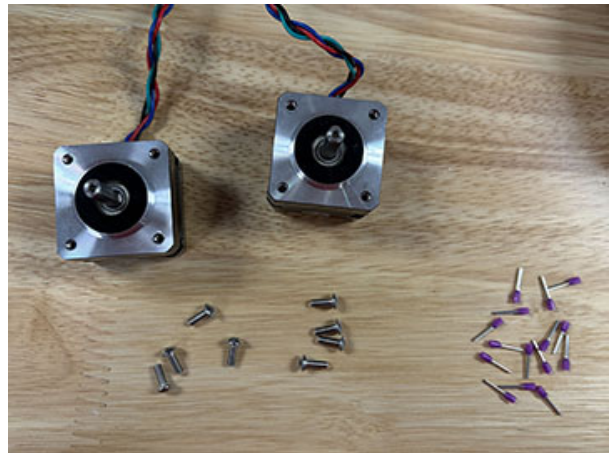


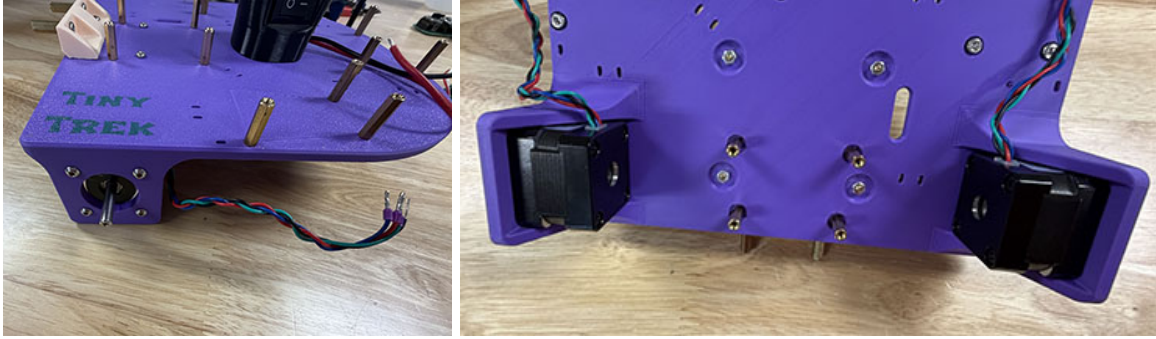
Using 3 M3x6mm screws each, install Motor Controller Boards on the two trapezoids. Note: There will be one unpopulated screw hole on each board. The screw hole is uncomfortably close to the pads for the CAN Bus so I explicitly didn't include the hole in the chassis to prevent accidentally shorting the CAN Bus with a standoff or screw.

Motors

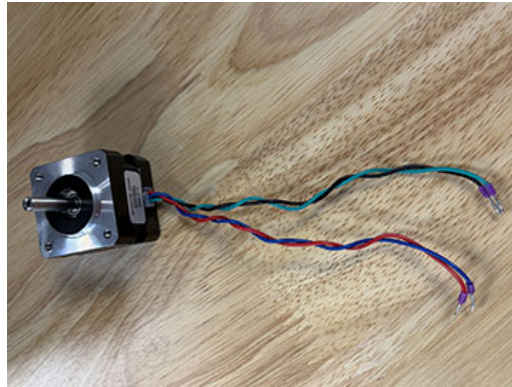
Parts:

- 2 x Nema 14 stepper Motors
- 8 x M3x8mm screws





Using 4 M3x8mm screws each, install motors with the wires pointing towards the back of the chassis.

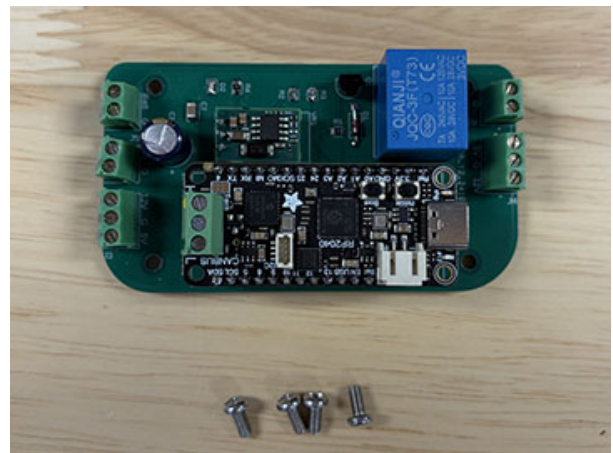


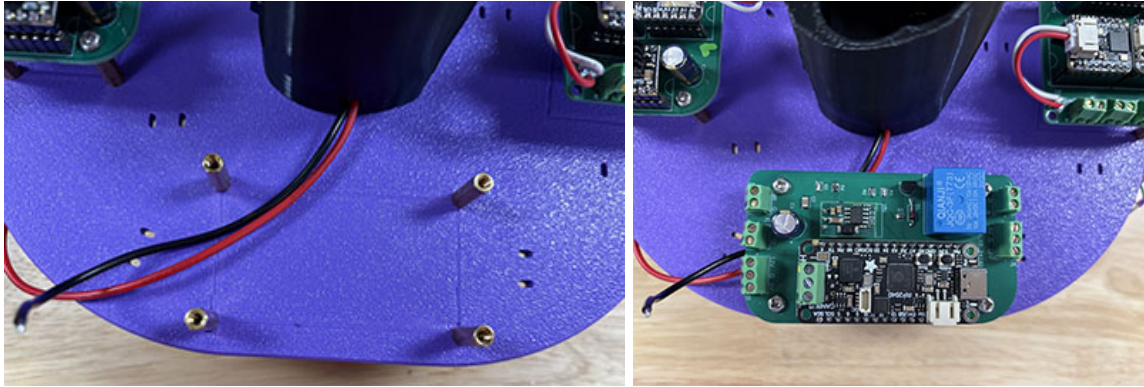
Optionally, use a drill to twist the two wires from each coil together. (i.e. red/blue and green/black) Cut wires to the appropriate length to reach the black screw terminal.

Power Distribution Board

Parts:

- 1 x Assembled Power Distribution Board
- 4 x M3x6mm screws



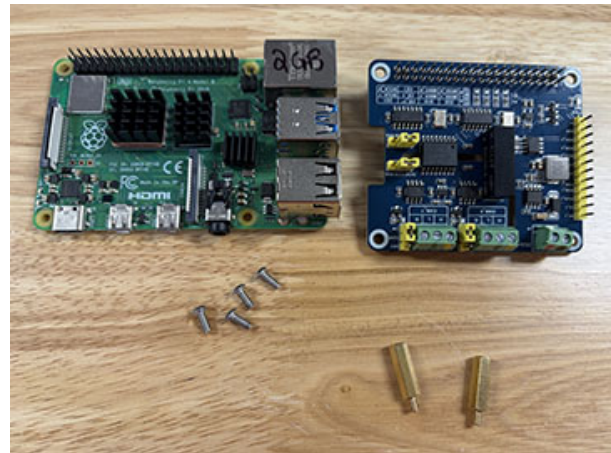


Route the wires from the battery receptacle between the standoffs as shown in the above left image. Use 4 M3x6mm screws to install the Power Distribution Board on the small rectangle with the Feather daughter board towards the back of the chassis.

Raspberry Pi and CAN FD Hat

Parts:

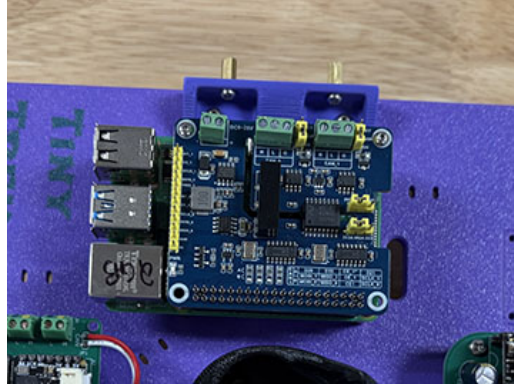
- 1 x Raspberry Pi 4B 2GB
- 1 x CAN FD Hat
- 4 x M2.5x6mm screws
- 2 x M2.5x15mm standoffs



BEFORE installing the Raspberry Pi, if you are using the Ultrasonic sensor, and/or the camera module, make sure the standoffs for the Ultrasonic Sensor and the Bracket for the Camera are installed. (see [Optional Ultrasonic Sensor and/or Camera Board](#))



Place the Raspberry Pi on the M2.5 standoffs. USB and Ethernet ports face the left side of the chassis. Use M2.5x6mm screws in the 2 holes on either side of the header bank, and M2.5x15mm standoffs in the front 2 holes.

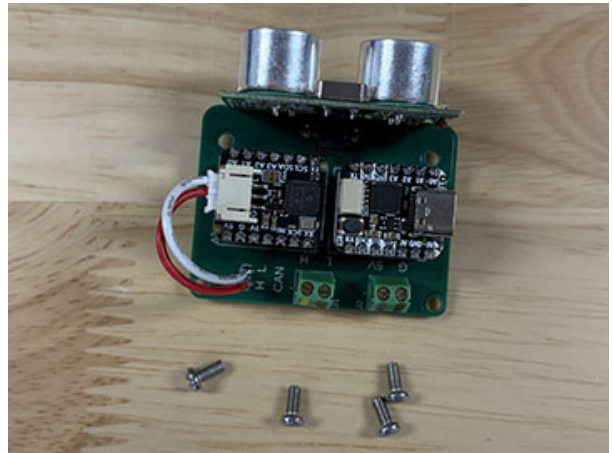


Install the CAN-FD Hat and screw down the front two holes with 2 more M2.5x6mm screws.

Ultrasonic Sensor Module

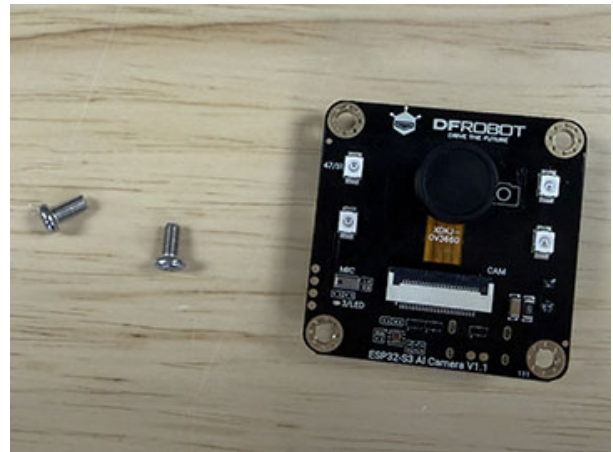
Parts:

- 1 x Assembled Ultrasonic Sensor Module
- 4 x M3x6mm screws



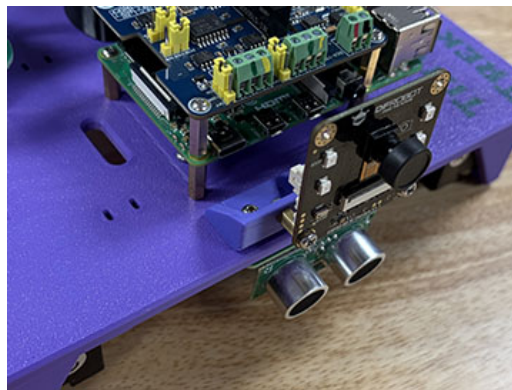
Install the Ultrasonic Board on the bottom of the chassis with 4 M3x6mm screws

Camera



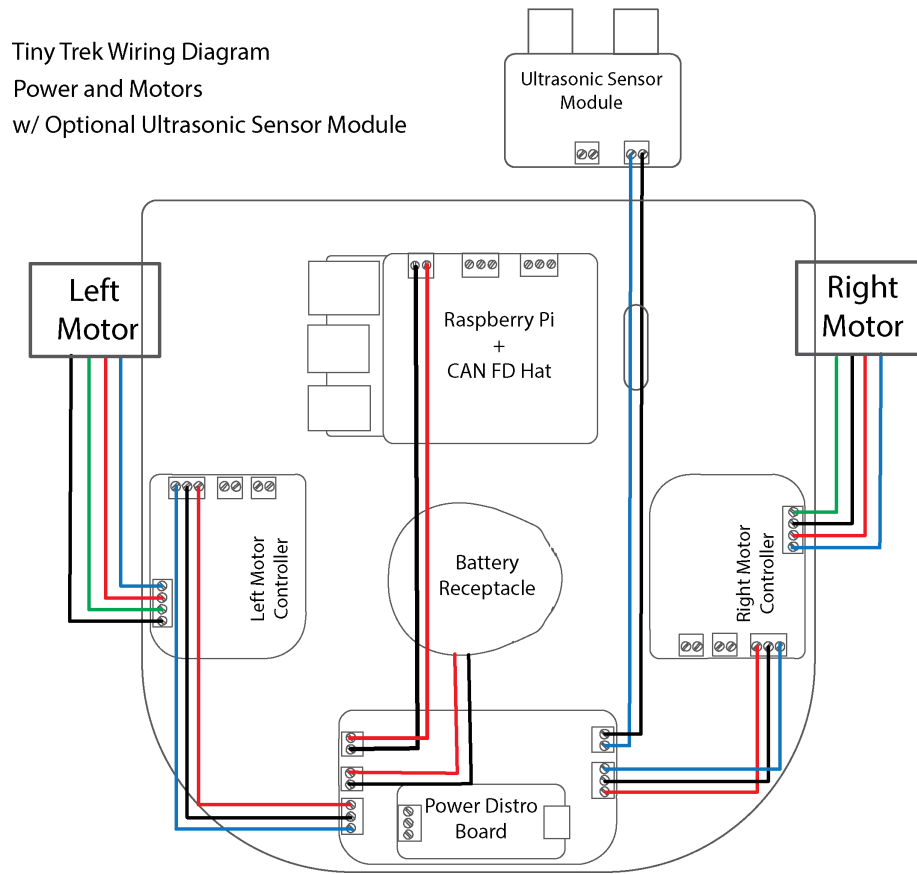
Parts:

- 1 x ESP32-S3 AI Camera Module
- 2 x M3x6mm screws



The camera board installs on the front of the camera bracket with 2 M3x6mm screws.

Wiring



I used Red wire for 12V and Blue for 5V. Optionally you could add a third color to differentiate between 12V straight from the battery and 12V out of the relay.

Set Motor Current Limit

Before you wire everything up, and before connecting the motors, you will want to set the current limit on the Motor Controller Modules. You will have to connect some of the wiring. Connect the battery to the Power Distribution Board. Connect 5V and Gnd from the Power Distribution Board to the Motor Controller and you will need 12V as well, without code to fire the Relay on the Power Distribution Board, you will need to hook up to the second port labeled Batt instead of 12V for this part. (This port ultimately powers the Raspberry Pi and you will need to hook up the Motor Control Module to the 12V line for normal use)

The current limit is calculated with the following (simplified) equation:

$$V_{REF} = 0.544\Omega \cdot I_{MAX}$$

In our case the current limit for our motor is $I_{MAX} = 0.5A/phase$

$$V_{REF} = 0.27V$$

If you keep the driver hard-wired to full-steps, the motor only uses 70% of the max current, so the max current could technically be set to 0.7A, thus $V_{REF} = 0.38V$.

The V_{REF} test point is indicated on the bottom of the motor driver by a white circle. V_{REF} can also be measured at the top of the potentiometer.

Motor Wiring

One coil (Blue/Red) 2B and 2A respectively on each motor controller module, the second coil (Green/Black) goes into 1A and 1B respectively on the left motor controller module, but 1B and 1A respectively on the right motor controller module. If the wheel turns in the wrong direction when testing, swap 1A and 1B on the corresponding motor controller module.

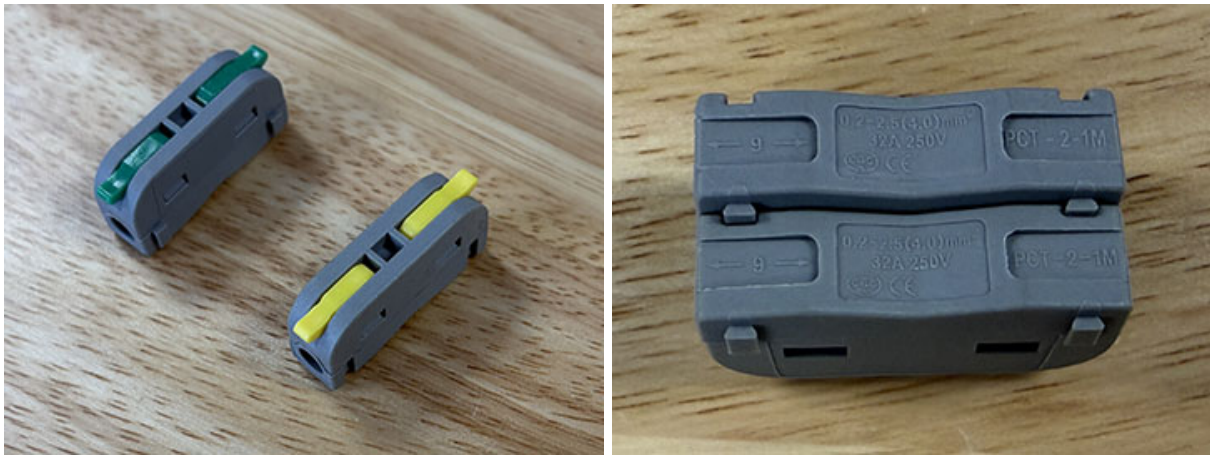
Ultrasonic Sensor Module

The large slot between the Raspberry Pi and the Snap Connectors is a pass-through for wires to reach the Ultrasonic Sensor Board on the bottom of the chassis.

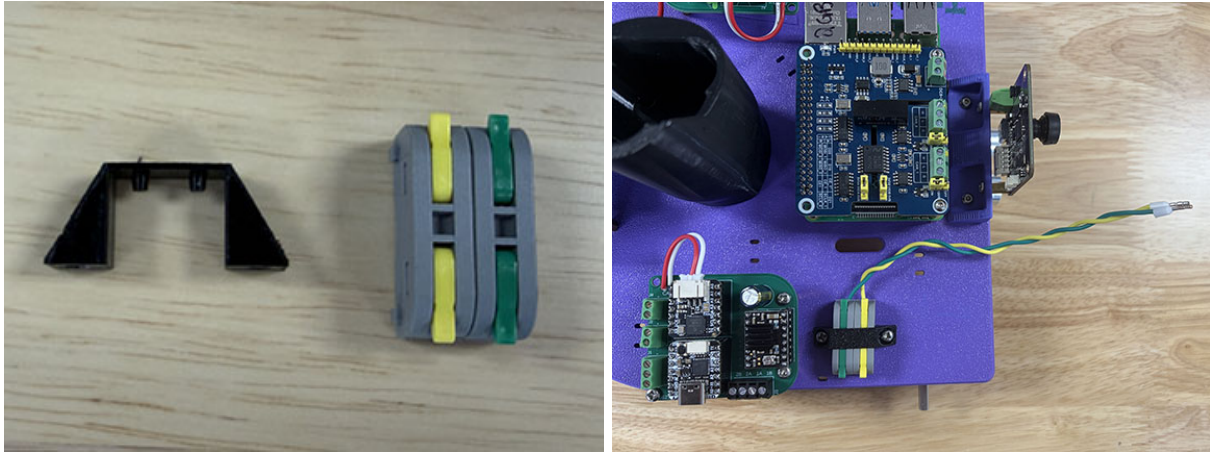
Ferules: OPTIONAL but highly recommended

I added ferules to the wires to help install wires into the screw terminals (And the Snap connectors). We're using 20AWG wire, but I had trouble fitting the 20AWG Ferules into the screw terminals. 22AWG ferules fit on the 20AWG wire and into the screw terminals I used. The motor wires are 26AWG so you will need 26AWG ferules for those wires. Any pre-tinned wires like the ones connected to the power switch should not need ferules attached.

Snap Connectors



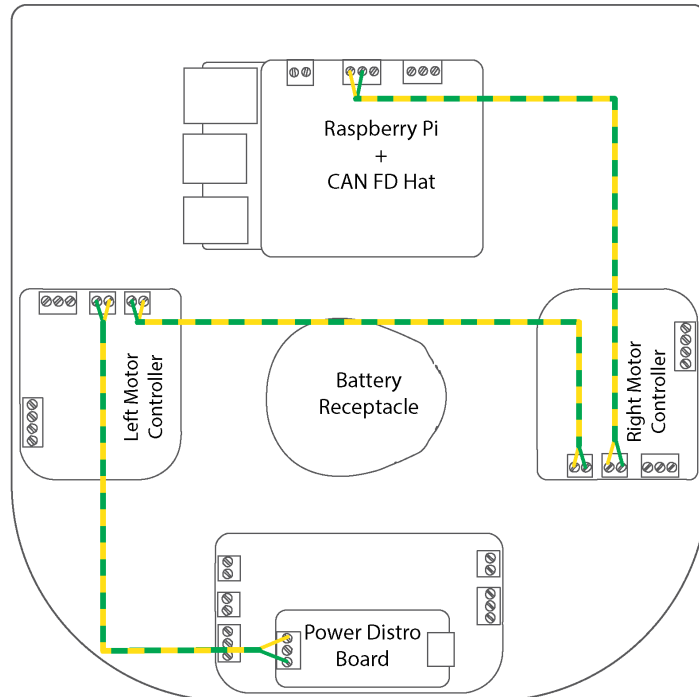
The snap connectors have a dovetail mating interface. Slot a Yellow connector and a Green connector together.



Slot the two nubs into the square holes on top of the snap connectors and screw the bracket down to the chassis with M3x6mm screws in the boxed pair of holes at the front right of the chassis.

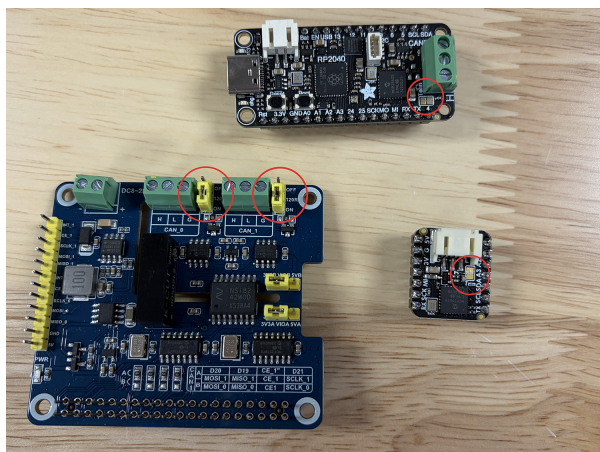
Given the space constraints, it may be easier to install a pair of CAN wires into one end of the Snap Connectors before installing the connectors on the chassis. These wires should face towards the Raspberry Pi when the Snap Connectors are installed.

Tiny Trek Wiring Diagram
CAN Bus



*Note: the CAN bus is a differential pair. This means the two wires should be as close to identical as possible and twisted together evenly along the full length with at least 1 twist per inch.

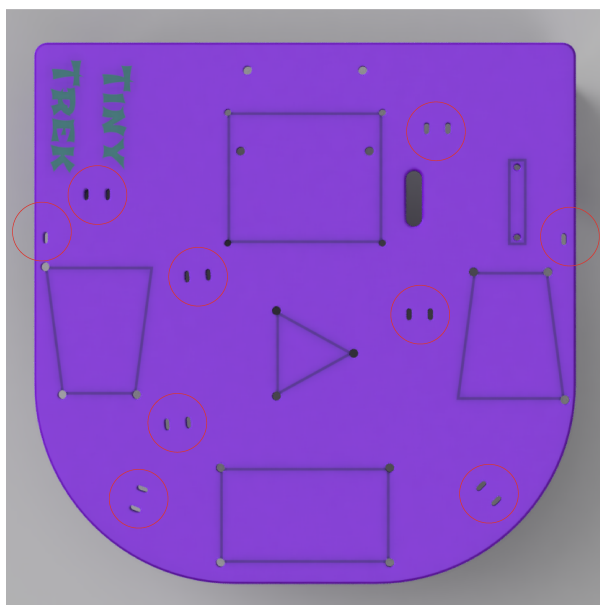
Terminating Resistors



The CAN Bus expects to have a 120Ω terminating resistor at each end of the bus. All of the CAN boards we used have terminating resistors installed, connected through a jumper circled in the photo above. In the above wiring configuration, the motor controllers are in the middle of the bus, so you have to cut the jumper on the top of the CAN BFF board. This can be done with a sharp hobby knife. The resistor on the Feather is also connected through a jumper with a cuttable trace. We want to leave it in this configuration, but you may want to cut it depending on your wiring. The resistors on the two buses on the CAN FD Hat are connected through physical jumpers on headers. In this configuration, both jumpers should be in the ON position.

Optional Ultrasonic Sensor:

Make sure to double check the jumper on the Ultrasonic Sensor Board is in the proper configuration for its location on the CAN bus. (not specified in this document)

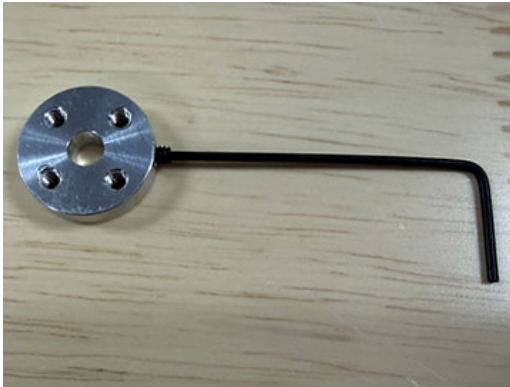


Once the wires have been routed, there are a few pairs of slots for wire management with 4" zipties. Near each motor there is a single slot. The ziptie is intended to wrap around the edge of the chassis here.

Wheel Assembly

Parts:

- 1 pair Pololu Wheels
- 1 pair Pololu Wheel Hubs
- 4 x M3x8mm screws



Install the set-screw in the wheel hub. Attach the hub to the wheel then install the assembly onto the motor shaft. The set-screw should line up with the flat side of the motor shaft.

A Note about Programming the Microcontrollers:

All three of the QTPYs (on the Motor Control Modules and the Ultrasonic Sensor Module) and the Feather are connected to the same 5V rail, so if you plug into their respective USB-C Ports, your computer will be trying to supply power to all four of these boards. The easiest way to make sure that doesn't happen is to remove the QTPY or Feather you're trying to program from the car entirely while you program it.